

NAJATH AKRAM, PH.D.

Houston, Texas | (330) 990-8406 | akram.m.n@ieee.org | linkedin.com/in/najath | najathakram.info

SUMMARY

Wireless systems engineer with ten years in radio, the last six taking LTE and 5G NR O-RAN radio units from architecture to commercial release. Works the seams between hardware, firmware, verification, and partner-facing teams: 3GPP and O-RAN alignment, interop debugging against OEM test stacks, and internal tools that lab teams run without instrument or MATLAB licenses. Owns the golden models firmware is signed off against at Airspan; promoted twice in four years to Principal at Jabil. Ships AI products end to end outside work and co-runs a wedding photography studio. Ph.D. with 10+ IEEE publications.

CORE SKILLS

Product and systems: radio unit architecture definition, 3GPP and O-RAN roadmap alignment, requirements and trade studies (massive MIMO, functional splits), verification strategy, release sign-off

Wireless depth: 5G NR and LTE PHY, O-RAN fronthaul (split 7.2x, WG4 CUS-plane, BFP compression), DFE, CFR, PRACH, EVM, ACLR, OBUE and PIM analysis, multiband TDD and FDD radios

Interop and delivery: OEM lab interop (Ericsson and VIAVI test stacks), field debug and root cause, cross-vendor O-RAN captures, internal tool delivery

AI and software: Anthropic Claude API and Agent SDK, multi-agent pipelines, cost governance, full-stack delivery (TypeScript, Python, MATLAB)

EXPERIENCE

AIRSPAN NETWORKS

Houston, Texas

Signal Processing Engineer

January 2025 to present

- Build and maintain the bit-accurate golden models of the downlink, uplink, low-PHY, and PRACH chains that firmware is signed off against for the multiband LTE and 5G NR O-RAN radio unit line: 3 to 100 MHz, FDD and TDD, combined-band carriers, O-RAN BFP at 8, 9, 10, 12, and 14 bits; adapted for an ODM partner's radio variant.
- Run Baseband Interface Unit modeling and feature bring-up across commercial releases.
- Ran production debug engagements, including interop against Ericsson and VIAVI test stacks, and traced lab and field captures to the root causes that unblocked releases.
- Shipped two standalone tools lab teams run without instrument or MATLAB licenses: an O-RAN fronthaul power and PAPR tool (MATLAB Compiler GUI and CLI executables plus a Python version) and an uplink EVM and PRACH analyzer aligned to the Keysight 89600 VSA, whose RU to DU full-scale alignment check separates radio misconfiguration from DU ingest error.
- Met an operator's 3 MHz carrier requirement by re-clocking CFR from 491.52 to 245.76 MSPS and redesigning the cancellation pulses and channel filters for 3 to 20 MHz (a 3 MHz channel filter is not realizable at 30.72 MSPS), and held 2.17 to 2.23% EVM inside ACLR and OBUE margin.
- Built a multiband PIM planner that flags IM3, IM5, and IM7 products landing in uplink bands before a band combination is committed to a filter plan.

JABIL INC.

Warren, New Jersey and Houston, Texas

Principal Wireless Systems Design Engineer

December 2023 to August 2024

Lead Wireless Systems Design Engineer

September 2021 to December 2023

Senior FPGA Design Engineer

December 2020 to September 2021

- Led the software and firmware workstreams for LTE and 5G NR O-RAN radio units from single-band to tri-band for OEM customers: defined NR implementation requirements and wrote the architecture and implementation documents the digital design team built to.
- Primary author of the DFE System Design Documents for two radio lines, a B3 FDD radio (5 to 40 MHz) and a 100 MHz TDD radio, and of the firmware implementation guides for NR phase compensation (TS 38.211 section 5.4) and RFSoc pre-emphasis.
- Wrote the uplink DFE receiver test plan for LTE, NR, and MSR against TS 36.141 and TS 38.141, from reference sensitivity through blocking.
- Weighed massive MIMO and O-RAN functional-split options against implementation complexity, resources, and timelines, and briefed customers on passive intermodulation in multiband radios.
- Delivered CFR cancellation-pulse analysis to a major OpenRAN program partner in 2022; ran filter design, CFR simulation, and end-to-end system modeling for single, dual, and tri-band radios through commercial release.

FLORIDA INTERNATIONAL UNIVERSITY

Graduate Research Assistant

Miami, Florida

2018 to 2020

- Developed digital and mixed-domain hardware-reduction algorithms for massive MIMO receivers: the basis of the dissertation and most of the 10+ IEEE publications, all on reduced-ADC array architectures.
- Implemented a 28 GHz digital beamforming array receiver with 800 MHz per channel on Xilinx RFSoc (ZCU111, ZCU1275, ZCU1285), cutting ADC count by 75% with frequency multiplexing, and showed a further 50% ADC reduction for 2D phased arrays over the air.

MATHWORKS

Engineering Intern

Natick, Massachusetts

2019

- Enhanced HDL Verifier workflows and developed Ethernet PHY-interface IP across Artix, Kintex, and Virtex FPGAs; two awards in the fall 2019 intern hackathon.

THE UNIVERSITY OF AKRON

Graduate Research Assistant

Akron, Ohio

2016 to 2018

- DARPA, NSF, and AFRL-funded research on multidimensional RF systems: approximate-DFT beamforming cores and multidimensional delta-sigma ADC architectures for phased-array receivers.

INDEPENDENT PRODUCTS AND BUSINESS

- **TaxLens:** rebuilds a year of small-business books from raw bank and card statements into a locked ledger, financials, and tax worksheets (Schedule C, 1120-S, 1065, 1120). Multi-agent AI pipeline; each deduction carries an IRS citation, an evidence rating, and a confidence score; on one real client year it caught \$10K of double-counted bounced payments that manual review had missed. It does not file. Anthropic API, Next.js, PostgreSQL.
- **Salient:** USMLE exam practice with an AI patient that never gives away the diagnosis: pre-authored case data so the model cannot invent a result, a leak guard that blocks answers that would reveal the diagnosis, a per-call cost ledger with daily budgets, and grading aligned with published exam rubrics. Live with Stripe subscriptions.
- **RouteFlow:** multi-tenant delivery and route-management SaaS for wholesale distributors, in production on web, mobile, and API; designed the tenancy, pricing, and deployment model. routeflow.info
- **Pixel Parchment:** co-owner of a Houston wedding photography studio. pixelparchment.com

EDUCATION

FLORIDA INTERNATIONAL UNIVERSITY Ph.D., Electrical and Computer Engineering, 2020

Miami, Florida

UNIVERSITY OF RUHUNA B.Sc. (Hons), Electrical and Information Engineering, 2016

Galle, Sri Lanka

NATIONAL INSTITUTE OF BUSINESS MANAGEMENT A.S., Business Management, 2015

Galle, Sri Lanka

Affiliations: IEEE, member since 2013; Tau Beta Pi Engineering Honor Society, 2019 to present

SELECTED PUBLICATIONS

- Frequency-Multiplexed Array Digitization for MIMO Receivers: 4-Antennas/ADC at 28 GHz on Xilinx ZCU-1285 RF SoC, IEEE Access, vol. 9, 2021.
- Spacetime Frequency-Multiplexed Digital-RF Array Receivers with Reduced ADC Count, IEEE Transactions on Circuits and Systems II: Express Briefs, vol. 68, 2021.
- Massive-MIMO and Digital mm-Wave Arrays on RF-SoCs using FDM for M-Fold Increase in Antennas per ADC/DAC, 2021 IEEE Space Hardware and Radio Conference (SHaRC), 2021.
- Fast Radix-32 Approximate DFTs for 1024-Beam Digital RF Beamforming, IEEE Access, vol. 8, 2020.
- A Direct-Conversion Digital Beamforming Array Receiver with 800 MHz Channel Bandwidth at 28 GHz using Xilinx RF SoC, IEEE COMCAS, Israel, 2019.
- Multiport ADCs for Microwave Focal Plane Array Dish Receivers, IEEE International Symposium on Circuits and Systems, Italy, 2018.

10+ IEEE publications, an AFRL technical report, and a dissertation; full list at najathakram.info.